

# Fourie van Rooyen

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## EDUCATION

The University of Alabama | Tuscaloosa, AL

Bachelor of Science in Aerospace Engineering | Expected May 2026 | Major GPA: 3.0

Relevant Coursework: Solid Mechanics, Aerospace Structures, High-Speed Aerodynamics, Propulsion, Thermodynamics

## RELEVANT SKILLS

Programming/Software: SolidWorks CAD, CATIA, ANSYS (CFD simulations), MATLAB, Python, Delphi 10, Jira

Hardware/Technical: CNC Mill, CNC Lathe, 3D Printing, GD&T (ASME Y14.5-2018)

## PROJECTS

### **Capstone Project, Tuscaloosa, AL, Chief Engineer**

**August 2025 – Present**

- Led system-level design of a fixed-wing eVTOL reconnaissance aircraft, integrating distributed electric propulsion, solar-film power augmentation, and modular ISR payloads.
- Defined mission requirements, Level I/II technical specs, and concept of operations (ConOps) in accordance with NASA MCR criteria and AIAA standards.
- Conducted aerodynamic sizing and performance analysis (takeoff, climb, cruise, loiter, and endurance) using CFD and analytical methods to inform configuration selection.
- Performed propulsion sizing, weight and balance calculations, structural load estimation, and control-surface definition to support stability and control requirements.
- Oversaw system architecture development including power distribution, sensor integration, avionics layout, and redundancy strategies to meet mission reliability targets.
- Directed risk analysis and preliminary SEMP/ICD development to support baseline design reviews and design verification planning.

### **Alabama Aerodynamics Experimental Research Operations (A.A.E.R.O.)**

#### **Co-Founder / CFD and Testing Lead**

**January 2025 – Present**

- Lead design of drone propeller with vortex inducers to evaluate aerodynamic performance.
- Simulated propeller thrust and acoustic characteristics using Computational Fluid dynamics (CFD) in ANSYS and SolidWorks and compared with real-world test data.
- Performed Finite Element Analysis (FEA) on designs to assess structural integrity and material performance.
- Applied Geometric Dimensioning and Tolerancing (GD&T) for precise manufacturing of components.
- Conducted static test stand experiments measuring thrust, RPM, and acoustics for multiple design iterations.
- Utilized wind tunnel for aerodynamic data collection and model validation.

### **Alabama Rocketry Association – Liquids Team Member**

**October 2023 – Present**

- Collaborated on designing components for a single-stage liquid propellant rocket, focusing on fluids systems.
- Designed test stand and created P&ID diagrams adhering to safety standards.
- Performed hydrostatic proof testing of flight hardware to demonstrate tank compliance with MIL-STD-810G/SMC-S-016 qualification requirements.
- Utilized a CNC Lathe to manufacture a graphite nozzle within design tolerances.
- Assisted in setup and execution of multiple hot fire tests for LOX/Kerosene engine, including short-duration and duration tests at Tuscaloosa National Airport, analyzing performance data to identify areas for improvement.

### **Genesis Drone Simulation Project, Personal Project**

**August 2025 - Present**

- Engineered a quadcopter simulation in Python using the Genesis physics engine with real-time aerodynamics, rigid-body dynamics, and control systems.
- Integrated PID controllers for altitude hold, attitude stabilization (roll, pitch, yaw), and position maintenance, tuned for critical damping ( $\zeta \approx 0.7-1.0$ ) and sub-second settling times.
- Implemented manual flight controls via Pygame with differential RPM mapping and automated hover/leveling using quaternion-to-Euler conversions.
- Built multi-environment support for parallel drone instances (>1000 steps/s on CUDA) with custom camera rendering, HUD overlays, and thrust/drag visualization.
- Optimized physical modeling with gravitational forces, empirical thrust/drag, and PID anti-windup to improve simulation fidelity and control accuracy.

## WORK EXPERIENCE

### **The Cube, Tuscaloosa, AL, Lab Technician**

**May 2022 – January 2024**

- Operate and maintain multiple 3D printers (FDM/FFF and composite systems) to support academic and research projects.
- Assist students and faculty with CAD file preparation, slicing, printing, and post-processing for rapid prototyping.
- Calibrate and troubleshoot equipment to ensure print quality and optimize build parameters.
- Enforce lab safety and equipment use protocols while training users on best practices.